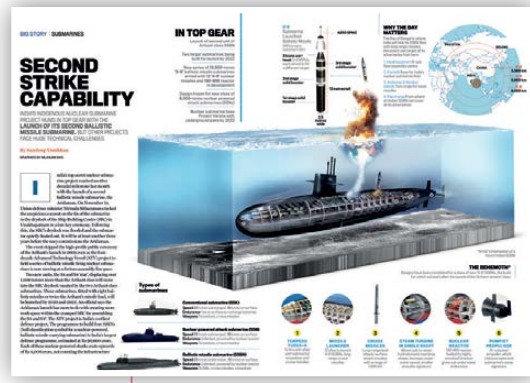

BIG STORY | SUBMARINES

SECOND STRIKE CAPABILITY

INDIA'S INDIGENOUS NUCLEAR SUBMARINE PROJECT HUMS IN TOP GEAR WITH THE LAUNCH OF ITS SECOND BALLISTIC MISSILE SUBMARINE. BUT OTHER PROJECTS FACE HUGE TECHNICAL CHALLENGES

By Sandeep Unnithan

IN TOP GEAR

WHY THE BAY MATTERS

Types of submarines

THE DOMINO EFFECT

FROM THE ARCHIVES

Dec. 18, 2017

Big Story **Second Strike Capability**

In 2006, a high-level committee under Dr R. Chidambaram, principal scientific advisor to the government of India, assessed India's ability to design and construct a class of three new SSBNs–the 'S5' class–to be fielded beginning in 2021. It budgeted Rs 10,000 crore, to be divided among BARC, DRDO and the ATV project headquarters, to begin the project by 2015

-Sandeep Unnithan

DID YOU KNOW?

The Advanced Technology Vessel programme was influenced by the USSR sending a nuclear submarine to successfully ward off the threat from US carrier USS *Enterprise* during the 1971 India–Pakistan war

signals progress in indigenous shipbuilding, nuclear propulsion and missile systems, thus reducing dependency on foreign suppliers.

Most importantly, *Arihant* satisfactorily fulfils India's strategic nuclear deterrent under the triad doctrine, which encompasses land, air and sea-based nuclear delivery systems. *Arihant* thus establishes another credible second-strike platform, ensuring India's ability to retaliate in the event of a nuclear-first strike. The stealth and mobility of Nuclear-powered Ballistic Missile Submarines (SSBNs) also make them nearly invulnerable to detection, solidifying India's deterrence posture.

The Arihant-class programme is part of the Advanced Technology Vessel (ATV) project initiated in the 1980s. With technical assistance from Russia, India developed a nuclear reactor capable of propelling submarines, ensuring sustained underwater operations for extended durations. INS *Arihant* has four vertical launch tubes that can carry 12 K-15 'Sagarika' missiles or four K-4 submarine launched ballistic missiles (SLBMs) that can carry nuclear warheads, with a range of 750 km and 3,500 km respectively. The submarine is powered by an 83 MW pressurised light-water reactor with enriched uranium fuel.

The submarine's operational reach extends into critical chokepoints like the Strait of Malacca and the South China Sea, allowing India to project power and counter Chinese naval presence. Given China's growing SSBN fleet and its increasing forays into the Indian Ocean Region (IOR), *Arihant* provides a crucial counterbalance. Actually, India's SSBN fleet is likely to trigger strategic recalibrations among neighbouring powers, with Pakistan hinting at developing its own sea-based deterrent and China further expanding its naval presence in the IOR.

While INS *Arihant* is a significant achievement, India faces challenges in operationalising its nuclear submarine programme. Limitations in submarine production capacity, reliance on imported technology for propulsion and sensors, and delays in missile development constrain its effectiveness. Furthermore, questions remain about crew training and integration into broader naval operations.

INS *Arihant* symbolises India's strategic aspirations and growing maritime influence. However, sustained investment in infrastructure, technology, and strategic planning is essential to maximise the potential of India's nuclear submarine programme. As regional power dynamics shift, *Arihant* and its successors will be crucial in safeguarding India's interests in an increasingly contested Indo-Pacific theatre. ■

UNDERWATER RIPOSTE

» **The other Arihant-class SSBN in service, INS *Arighat*, and the under-construction *Aridaman*, will enhance second-strike capability**

» **An SSBN allows India to deploy its nuclear assets far from its shores**

» **India is developing nuclear-powered attack submarines (SSNs) to sharpen conventional naval dominance and anti-submarine warfare**

» **It complements India's strategic partnerships with US, Japan and Australia under the Quad**

» **With Pakistan exploring sea-based nuclear deterrents, *Arihant* provides a technological and operational edge**